

HW 1

Excel block

1. (10 pts) You are given the data on Customers in the "Customer.xlsx" file. Your task is to create a new list and on it you do the following:

a. Display the number of times you meet the customers with the following names: Jay, Jeremy, Sheryl, Andrew and Bob. The cells where the respective number of occurrences are to be displayed are A1, A2, A3, A4 and A5

b. Display the number of rows in the database where at least one of the conditions is met:

b1. The person's first name starts with "J" and their email ends with ".com"

b2. The customer's corresponding city has more than 10 characters as does his country (spaces are included)

b3. The customer's first and last name start with the letter K

c. Build a histogram with the frequency of the random 5 names in the database (you can choose the names)

Python block

2. (10 pts) You have two matrices that are represented as lists of lists (Ex: [[1,2,3], [4,5,6], [7,8,9]] is a 3x3 matrix) or vectors that are represented as python lists. Your task is to implement a function, which takes 2 arguments as input (should be represented as 2 input variables) and returns the matrix/ vector, where each element is the sum of the elements with corresponding indices in the two input variables. Your function must handle the errors that could happen during the computation. In the case of error it should print "An error has occurred during computation" and return the None value. Name this function "elementwise_mv_add"

3. (20 pts) Create a phone book with a layered menu (I mean you could also do one layer, but pls don't). Your script should be able to:

1. Display the current entries in the book
2. Create a new entry with data input by the user
3. Delete an existing entry
4. Change an existing entry
5. The program should retain knowledge of the entries even after restarting
6. Handle errors in the entry refactoring/ addition/ deletion/ fileread by either asking user to input the information again or displaying a text that notifies of the error (not any specific error, just about the fact that there's an error in the program) and shutting down
7. Show the number of entries in the database

Bonus task

4. (5 pts) Write a generator that replicates the python “range” function with a twist. It now should print the floating point values, all arguments are the same